

From Files to Trusted Snowflake Data Products

Inspect, load, transform, reconcile, and publish.

CSV

explicit parsing

JSON

preserved shape

PARQUET

embedded schema

AI-Free Zone: reason, write, validate, and explain your own work.

A working query is the beginning, not the finish line

SOURCE

Did we read the
exact file?

01

GRAIN

What does one
row represent?

02

RULES

Why did rows
change?

03

EVIDENCE

Can we prove
the result?

04

Trust is built through observable decisions.

Today you will build, validate, and defend a small pipeline

- 1 Convert messy text safely and classify quality
- 2 Inspect staged files before loading
- 3 Load CSV and JSON with explicit contracts
- 4 Reuse the January Parquet fact table
- 5 Build clean, rejected, mart, and view layers
- 6 **Reconcile counts and defend design trade-offs**

The outcome is a pipeline another engineer can inspect and rerun.

Three formats carry different promises and different risks

CSV

Rules live outside the values

delimiter, header, quotes

Risk: silent column shift

JSON

Names and nested shape travel with values

keys, paths, object shape

Risk: changing types or keys

PARQUET

Column names and types are embedded

schema, typed columns, period

Risk: valid structure, wrong data

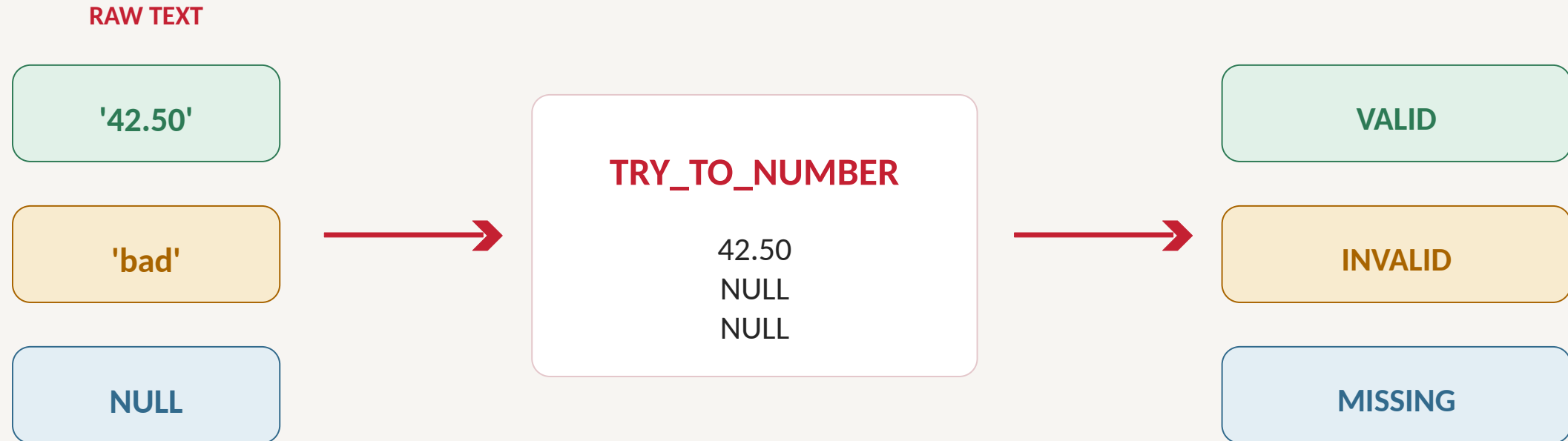
A file extension is a clue. A tested contract is evidence.

Snowflake separates location, interpretation, movement, and proof



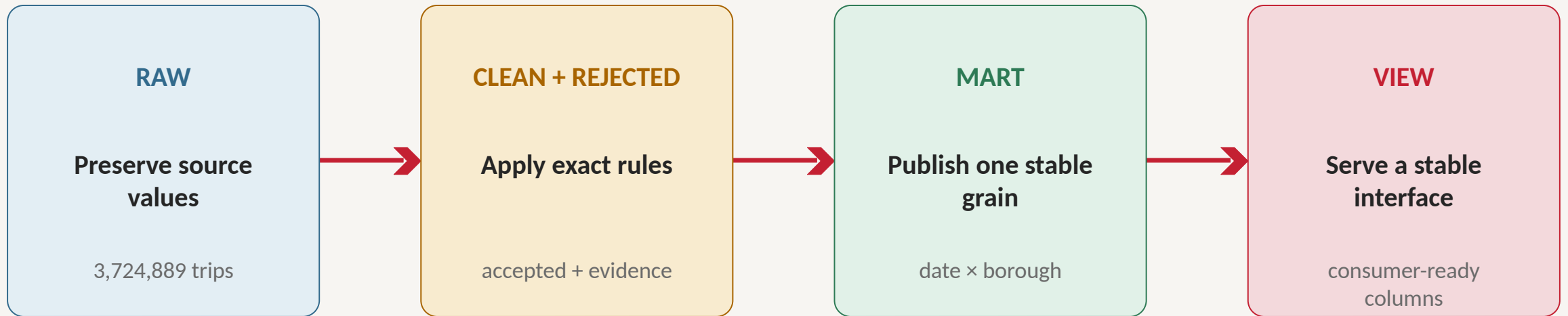
Keep cloud keys out of learner code. Use the approved storage integration.

Safe conversion turns an exception into measurable evidence



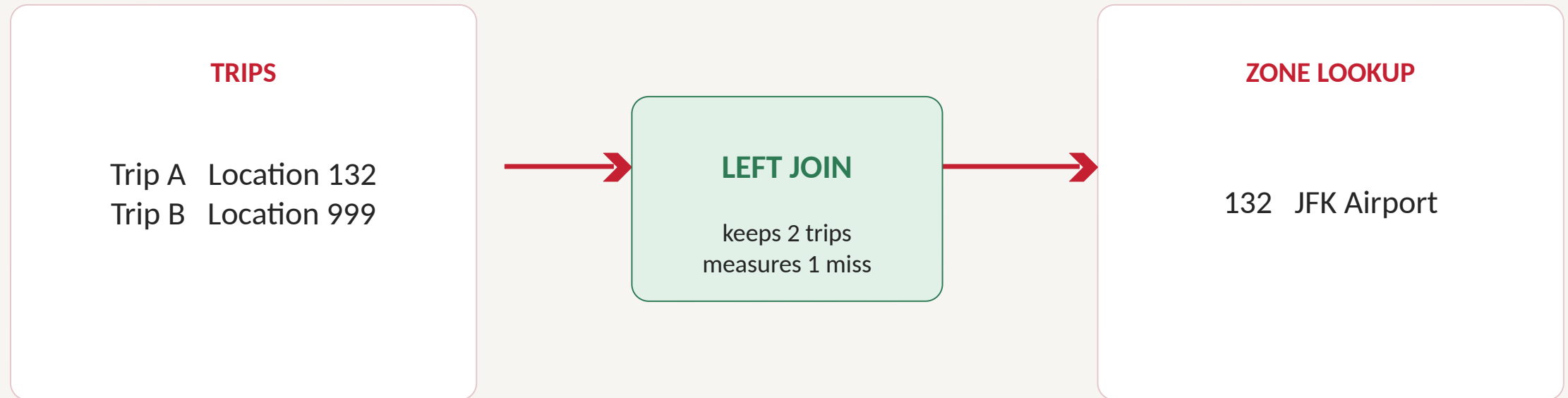
NULL is an outcome. Quality status is the decision evidence.

Separate layers make every population change explainable



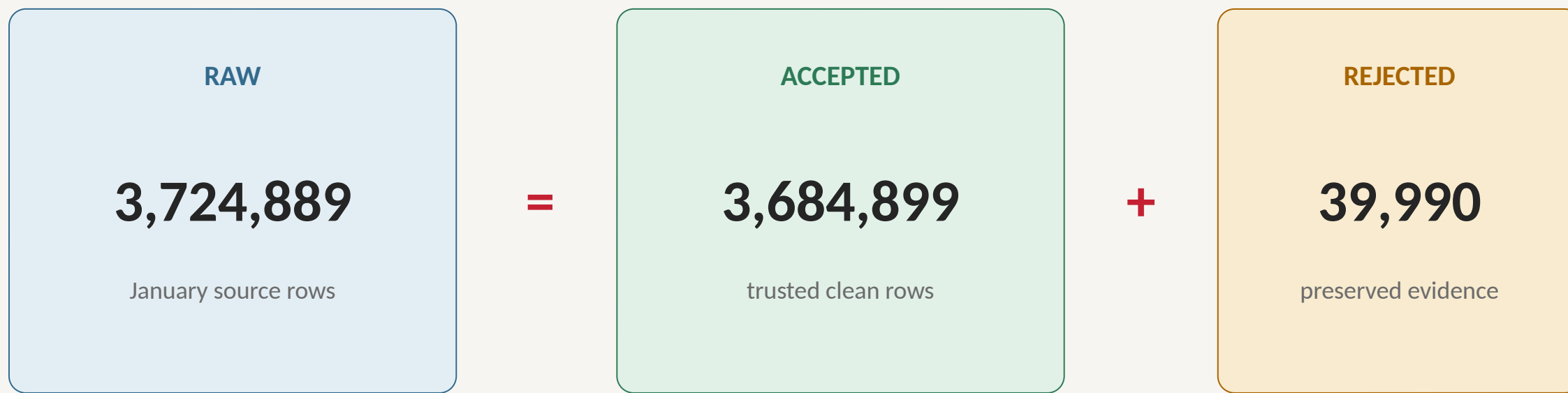
When a metric changes, ask which layer changed the meaning.

Preserve the fact rows before deciding how to handle missing keys



An INNER JOIN can make a missing-key problem disappear from the result.

Every raw row must have an explained destination



Expected mart: 31 service dates, 248 date × borough rows, 0 unmatched zones.

Choose CTAS, INSERT, or a view based on lifecycle and rerun needs

CTAS

Build a materialized result

Simple full refresh

Rerun by rebuilding

INSERT INTO

Append a controlled batch

Incremental pattern

Requires duplicate handling

VIEW

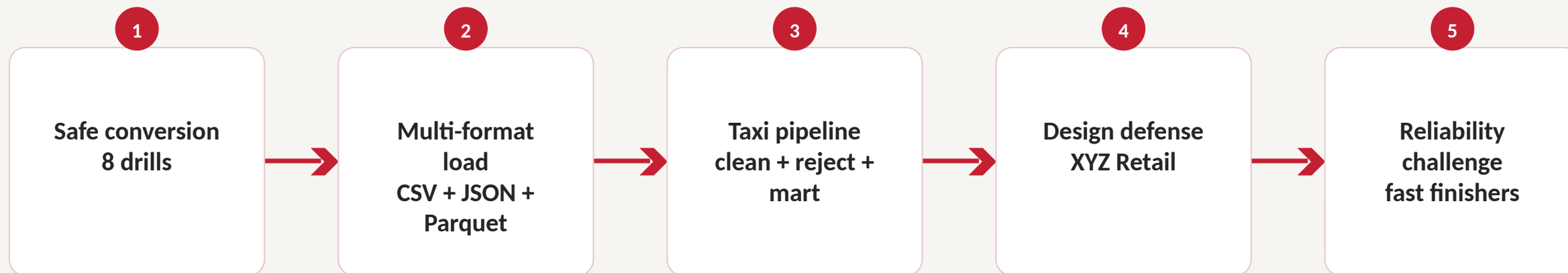
Store query logic

Stable consumer interface

Reads current sources

First make the manual run trustworthy. Then automate the right behavior.

The day moves from small drills to a complete, defended system



Required evidence

load results • row counts • grain tests • reconciliation • design trade-off

A trustworthy pipeline can answer five questions without guesswork

- 1 Which exact file did we read?
- 2 What does one row mean?
- 3 Which rules changed the population?
- 4 What happens after a failure?
- 5 **What evidence proves publication is safe?**

Draft today
Defend on Week 5 Day 1

Close Week 4 by making every decision visible.